

Using MRI for Brain Tumor Detection and Segmentation

Isha Dev

Department of Biomedical Engineering
Johns Hopkins University Whiting School of
Engineering
Baltimore, Maryland, USA
idev1@jh.edu

Dharvi Jagirdar

Department of Biomedical Engineering
Johns Hopkins University Whiting School of
Engineering
Baltimore, Maryland, USA
djagird1@jh.edu

Rohan Giridharan

Department of Electrical and Computer
Engineering
Johns Hopkins University Whiting School of
Engineering
Baltimore, Maryland, USA
rgiridh1@jh.edu

Abstract— This work presents a multimodal pipeline for brain tumor segmentation and clinical interpretation using MRI, combining segmentation, synthetic data augmentation and large language model (LLM) based reporting. Utilizing the BraTS 2020 dataset, an Attention U-Net (AU-Net) segments tumor subregions across four MRI modalities. A Denoising Diffusion Probabilistic Model (DDPM) generates synthetic MRI volumes to address data imbalance and boost model generalization. The final segmentation outputs are used to create structured textual and visual prompts that guide the Radiology Infer-Mini LLM in producing clinical-style reports. The segmentation model achieved high accuracy, sensitivity and mIoU with synthetic augmentation, and the generated reports attained an average embedding similarity of 0.35.

Keywords— Brain tumor segmentation, Attention U-Net, diffusion models, synthetic medical imaging, radiology LLMs, multimodal MRI, BraTS dataset.

I. INTRODUCTION

Brain tumor diagnosis and treatment planning depend heavily on accurate interpretation of Magnetic Resonance Imaging (MRI) scans, a process that is time-intensive, subjective and dependent on radiologist and physician expertise. Current works on deep-learning models to segment brain tumors using Magnetic Resonance Imaging (MRI) data primarily utilize single-modality pipelines with U-Net and CNN architectures [1]. However, these methods do not inherently convey clinical insights necessary for prognosis and intervention planning. While some studies use multi-modal MRI data to enhance segmentation performance, limited studies integrate attention-based mechanisms, address annotation bias and data imbalance, and strive to improve interpretability and clinical implementation [2][3]. Furthermore, even fewer link segmentation to severity and prognostic analyses.

Integrating LLMs with medical image segmentation not only enhances interpretability, but also has the potential to streamline tumor reviews, reduce diagnostic errors, and improve patient outcomes. If successful, this work could be a small step towards redefining how segmentation outputs are communicated.

The core problem addressed in this project is the lack of diverse brain tumor MRI data, interpretability and clinical usability in current brain tumor segmentation pipelines. Specifically, the aim is to improve current segmentation pipelines and bridge the gap between segmentation outputs and clinical interpretability and reporting. The scope of this work is focused on glioma detection using the BraTS 2020 dataset. It demonstrates a generalized approach for multimodal MRI interpretation that can be extended to external data and tasks. By combining segmentation outputs with a domain-specific LLM, this work introduces an end-to-end framework that has the potential to not only segment tumor subregions but also explain them in a clinically relevant format.

II. BACKGROUND

To enhance brain tumor segmentation and classification, a combination of advanced techniques was employed. Attention U-Net (AU-Net) extends the traditional U-Net by incorporating attention gates that emphasize relevant tumor-specific information while suppressing irrelevant features. Particularly, the AU-Net uses compressed encoder features and decoder features which are added element wise during attention activation. This mechanism compares encoder, and decoder features and applies ReLU activation to focus on relevant features being upsampled in the decoder. A Denoising Diffusion Probabilistic Model (DDPM) generates synthetic MRI data by iteratively reversing a noise-adding process, addressing medical image data sparsity. The Radiology-Infer-Mini is a domain-specific LLM model which was used to create clinically coherent reports for interpretability.

III. MATERIALS AND METHODS

A. Materials

To solve the issue of class imbalance, the Brain Tumor Segmentation Challenge (BraTS) 2020 dataset was used [4]. This dataset is a publicly available, expertly curated benchmark dataset designed to evaluate algorithms for brain tumor detection and subregion segmentation using multimodal scans. It focuses on patient diagnosed with gliomas and includes MRI scans collected from different institutions using different

scanners and imaging protocols. Each subject includes four 3D modalities – T1, T1ce, T2 and FLAIR, as well as voxel-wise ground truth annotations segmented by expert neuroradiologists into four classes – background, necrotic core, peritumoral edema and enhancing tumor.

B. Methodology

The first stage of the pipeline employed a Denoising Diffusion Probabilistic Model (DDPM) [5] to generate high-quality synthetic training images across all five MRI modalities (FLAIR, T1, T1CE, T2, and segmentation labels), ensuring balanced representations of tumor and non-tumor regions. These synthetic images were combined with the original dataset to train the U-Net segmentation model, enhancing its ability to recognize diverse tumor appearances. Initially, a Generative Adversarial Network (GAN) was explored for synthetic data generation. However, despite extensive experimentation with various GAN architectures, it consistently failed to produce high-quality images, suffering from instability, mode collapse, and high computational costs.

In contrast, the DDPM demonstrated superior stability, efficiency, and image quality. It employed a linear noise schedule with 200 diffusion steps, where Gaussian noise was gradually added to the images during the forward process. The model was trained to predict this noise, allowing for stable and efficient learning. During inference, the model generated images by reversing this process, starting from pure noise and iteratively refining it into detailed, realistic MRI images. This denoising process inherently avoided the adversarial training challenges of GANs and consistently produced anatomically accurate and detailed MRI images, including clear tumor representations.

The DDPM was implemented using TensorFlow with a high-performance data pipeline, which efficiently loaded and preprocessed multi-modal MRI images directly into RAM. Each MRI volume was normalized to the range $[-1, 1]$, and a custom function was used to dynamically sample random slices for training, ensuring data diversity. The model architecture was a 2D U-Net with sinusoidal timestep embeddings, consisting of downsampling, bottleneck, and upsampling blocks with Group Normalization and SiLU activations. Key performance optimizations included mixed precision training (XLA JIT, float16) and GPU memory growth management. Multi-scale Structural Similarity (MS-SSIM) was employed as an additional evaluation metric, assessing image quality across all modalities. Model checkpoints and synthetic outputs were saved at regular intervals, supporting efficient training, monitoring, and reproducibility.

U-Net is a convolutional neural network that has been developed specifically for biomedical image segmentation that

features a symmetric encoder-decoder structure where the contracting encoder captures and condenses image features through successive convolution and pooling, and the expanding decoder restores spatial resolution using up sampling and transposed convolution [6]. Skip connections merge high-resolution features from the encoder with the upsampled output of the decoder, enabling precise segmentation. However, this brings across redundant features, as the feature representation in the initial layers can be suboptimal.

To address this, attention gates are incorporated in the U-Net architecture at the skip connections to suppress irrelevant features, thereby improving the U-Net performance by highlighting relevant features in the images [7].

The U-Net model, adapted from an open-source Kaggle notebook that used the BraTS dataset [4][8], was implemented using TensorFlow and Keras to segment brain tumors from multimodal MRI scans. The model focused on the four MRI modalities T1, T2, T1ce and fluid-attenuated inversion recovery (FLAIR) images. The dataset was split into training (70%), validation (20%) and testing (10%) sets. Synthetic MRI data, generated using the DDPM technique, was incorporated to enhance model robustness and generalization.

Preprocessing steps included normalization and resizing of the images and a 1x1 convolution layer with a SoftMax activation function is used to predict multi-class tumor regions. A grid search was performed across different learning rates and batch sizes to identify the most optimal hyperparameters for the model. The best performance was achieved with a learning rate of 0.0003 and batch size of 4 conducted over 25 epochs. This model can generate detailed pixel-wise predictions and produce segmentation masks that delineate tumor regions from surrounding healthy tissue. Further, this model can distinguish between multiple tumor classes: necrotic core, edema and enhancing tumor regions [8].

The model was trained using categorical cross-entropy loss and optimized with the Adam optimizer. The model performance was evaluated using various metrics including accuracy, mean IoU, precision, specificity, sensitivity, overall dice coefficients and per class dice coefficient.

A vision-language generation pipeline was implemented using the Radiology Infer-Mini model [9], built on Qwen2L [10]. While a larger model, such as BioGPT, would have been better for this application, limitations were posed by the lack of reliability on Rockfish and the limited RAM and resources available on Google Colab. Radiology Infer-Mini, while fine-tuned for radiology-specific applications, is a lighter model that is limited in the amount of medical context it has been trained on. First, structured summaries were aggregated from the segmentation results - including tumor volumes, presence of subregions (core, edema, enhancing), anatomical locations, and

radiological features - into a prompt. This prompt was paired with two flattened montage images: one showing the T1ce images and the other showing model prediction segmentation.

The images and prompt were formatted into a multimodal message and passed to the LLM, which then generated a structured clinical report including sections on tumor overview, class-specific morphology, differential diagnosis, prognostic insights, and literature references. Outputs were collected for all unseen test subjects used in the segmentation model evaluation. Given the computational constraints, a limited range of temperature, repetition penalty and top-p values were tested using the embedding similarity metric. The final values used were 0.8 temperature, 1.2 repetition penalty and 0.9 top-p, which led to the greatest mean and maximum similarity values.

To assess factual alignment between prompts and generated outputs, the all-MiniLM-L6-v2 sentence transformer model was used to compute cosine similarity scores between the prompt and its corresponding LLM response. This embedding-based [10][12] similarity metric offered a lightweight, automated method to assess semantic fidelity.

IV. RESULTS AND DISCUSSION

The results of the DDPM implementation demonstrate its capacity to generate high-quality synthetic MRI images across all five modalities (FLAIR, T1, T1CE, T2, and segmentation labels). The model achieved an overall average Multi-Scale Structural Similarity Index Measure (MS-SSIM) of 0.5741, with individual modality averages ranging from 0.5440 for segmentation (SEG) to 0.5891 for T2. These values are consistent with the recommended MS-SSIM range of 0.3-0.7, which has been identified in recent research [11] as indicative of a balanced trade-off between maintaining fidelity to the original data and introducing sufficient structural diversity. This balance is critical for generating synthetic images that are both visually and anatomically accurate while also providing variability that can enhance downstream model training.

TABLE I. MS-SSIM values for synthetic MRI images generated by DDPM

Modality	Mean	Min	Max
Flair	0.5866	0.3669	0.7353
T1	0.5776	0.3657	0.7480
T1ce	0.5734	0.3455	0.7309
T2	0.5891	0.3711	0.7373
Seg	0.5440	0.3269	0.6919
Overall	0.5741	0.3560	0.7207

Qualitative evaluation further supports the efficacy of the DDPM-generated images, which preserve anatomically accurate structures, including clear tumor representations, well-defined brain anatomy, and consistent intensity distributions

across modalities. Such anatomical integrity is essential for clinical usability, particularly in the context of training a U-Net segmentation model. However, certain limitations are evident. The lowest MS-SSIM values, particularly for the segmentation labels (min=0.3269), suggest occasional deficiencies in fine-grained detail or minor misalignments in synthetic segmentations. These challenges could potentially be addressed through enhancements such as a more sophisticated noise schedule, architectural adjustments to the U-Net backbone, or the inclusion of a hybrid loss function that directly optimizes segmentation quality.

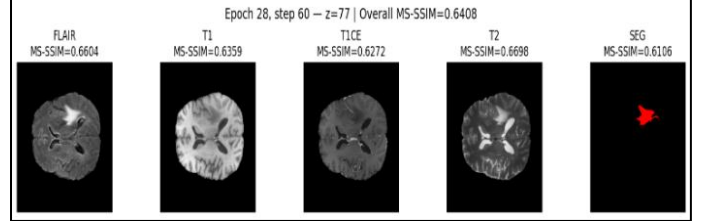


Fig 1. Sample synthetic MRI images

Although MS-SSIM was selected as the primary evaluation metric due to its sensitivity to structural information, the Fréchet Inception Distance (FID) was also considered. However, FID was ultimately not utilized due to resource constraints. Computing FID requires extensive memory and processing power because it relies on feature extraction using a pre-trained neural network, typically InceptionNet, followed by a comparison of the distributions between generated and real images. Implementing FID would have required training an additional InceptionNet model for feature extraction, which exceeded the available computational resources. Despite this limitation, the MS-SSIM values and qualitative assessments demonstrate that the DDPM effectively generated diverse, high-quality synthetic MRI data, contributing to a more robust and generalizable segmentation model.

TABLE II: Evaluation metrics for all model configurations

Metric	U-Net	AU-Net	AU-Net + Synthetic
Loss	0.0325	0.0303	0.0320
Accuracy	0.9884	0.9894	0.9887
MeanIOU	0.7648	0.7790	0.8204
Dice Coef	0.5248	0.5252	0.5157
Precision	0.9915	0.9925	0.9905
Sensitivity	0.9850	0.9853	0.9968
Specificity	0.9971	0.9974	0.9968
Dice (Necrotic)	0.4895	0.4537	0.3862
Dice (Edema)	0.5968	0.6730	0.6256
Dice (Enhancing)	0.5289	0.5609	0.5036

Three model configurations were evaluated on the BraTS 2020 dataset: a U-Net, an AU-Net and an AU-Net trained with synthetic data augmentation. As seen in Table 2, while all

models performed well in terms of pixel-wise classification accuracy (~99%), more granular analysis using Dice coefficients and mean intersection over union (mIoU) revealed important differences.

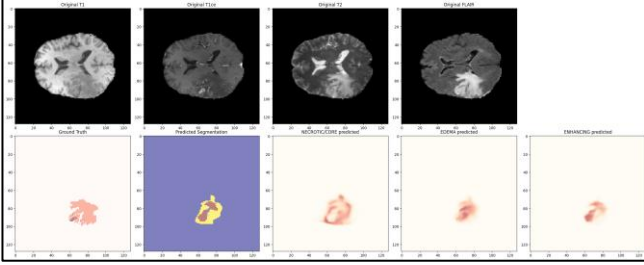


Fig 2. Sample output from AU-Net including ground truth and prediction

The AU-Net consistently outperformed the baseline U-Net on tumor subregion segmentation, particularly for edema and enhancing tumor regions that are typically difficult to diagnose, demonstrating the benefit of the attention gates. These gates helped the model focus on salient tumor features while suppressing irrelevant background voxels, thus improving the segmentation of clinically relevant regions. The AU-Net model trained with synthetic data achieved the highest mIoU and sensitivity among all three, suggesting that including DDPM-generated images during training helped the model generalize better and reduce false negatives. However, there was a drop in the Dice scores for the necrotic and enhancing classes. This could be a result of the subtle differences in the structure of synthetic samples compared to real data, leading the model to sometimes over or under-segment subregions.

TABLE III. Radiology Infer-Mini embedding similarity scores

Metric	AU-Net	AU-Net + Synthetic
No. Cases	33	44
Mean	0.3434	0.3499
Standard Deviation	0.0726	0.1070
Min	0.2325	0.2320
Max	0.5824	0.6975
Median	0.3488	0.3234

To bridge the gap between raw segmentation outputs and clinical interpretability, the Radiology Infer-Mini vision-language model was evaluated on test-time predictions from both AU-Net and AU-Net trained using synthetic data. The goal was to assess the LLM’s ability to translate segmentation-derived prompts into coherent narratives.

Embedding scores, shown in Table 3, demonstrate the both models led to a similar LLM performance with mean scores around 0.35. This suggests that while the LLM was able to extract some clinical information embedded in the structured prompts, there was a significant presence of hallucinations and variability. These are known issues in medical LLM usage with

real-world implications. Qualitatively, the generated reports mirrored real-world radiological language.

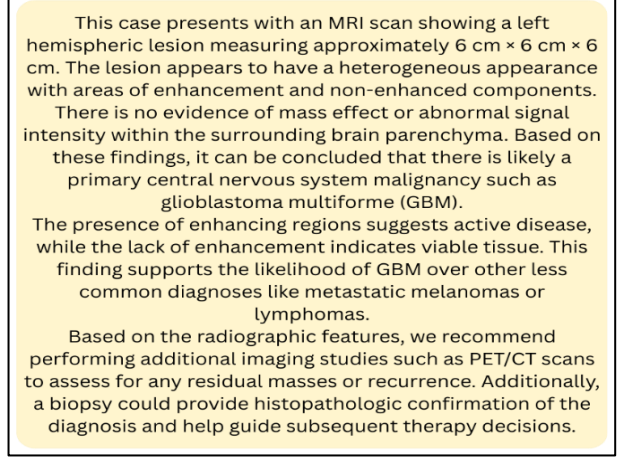


Fig 3. Sample Radiology Infer-Mini Output Report

In several cases, the LLM inferred treatment strategies such as surgical resection, chemotherapy, among others. The variance in scores emphasize the need for future LLM pipelines to include more robust validation and checks – such as retrieval augmented generation, or rule-based fact validation – to further improve reliability in clinical reporting tasks.

V. CONCLUSIONS

This project successfully demonstrates the advantage of integrating attention mechanisms and synthetic data augmentation for brain tumor detection and segmentation using multimodal MRI scans. The findings suggest that the addition of synthetic data can improve and boost model performance, generalization and sensitivity, however it may introduce slight variability in the segmentation. While the use of synthetic data may introduce minor inconsistencies in class-specific segmentation, it significantly enhances the model’s overall detection capability. This work offers a promising direction towards enhancing model robustness for medical image segmentation in cases of data sparsity. Future work may benefit from fine-tuning the balance of real versus synthetic samples and exploring techniques to better align synthetic data distributions with real-world variability.

This integration of vision-language models with segmentation pipelines represents a promising step towards generating automated and interpretable case summaries that can support clinical decision-making and multidisciplinary tumor reviews. Future work should include manual expert-led validation, stronger similarity metrics and rule-based fact validation.

Ultimately, this work serves as a useful baseline that demonstrates that combining attention mechanisms with synthetic data augmentation is a viable technique. With further research and optimization our work can realize its benefits in clinical and real-world applications.

REFERENCES

- [1] Liu, Z., Tong, L., Chen, L., Jiang, Z., Zhou, F., & Zhang, Q., Zhang, X., Jin, Y., Zhou, H. (2023). Deep learning for brain tumor segmentation: Current achievements and open challenges. *Complex & Intelligent Systems*. <https://doi.org/10.1007/s40747-022-00815-5>
- [2] Liu, Z., Tong, L., Chen, L., Jiang, Z., Zhou, F., Zhang, Q., Zhang, X., Jin, Y., & Zhou, H. (2022). Deep learning based brain tumor segmentation: a survey. *Complex & Intelligent Systems*, 9(1), 1001–1026. <https://doi.org/10.1007/s40747-022-00815-5>
- [3] Ahamed, M. F., Hossain, M. M., Nahiduzzaman, M., Islam, M. R., Islam, M. R., Ahsan, M., & Haider, J. (2023). A review on brain tumor segmentation based on deep learning methods with federated learning techniques. *Computerized Medical Imaging and Graphics*, 110, 102313. <https://doi.org/10.1016/j.compmedimag.2023.102313>
- [4] Center for Biomedical Image Computing & Analytics. (2020). Brain Tumor Segmentation (BraTS) Challenge 2020: Scope. Perelman School of Medicine at the University of Pennsylvania. <https://www.med.upenn.edu/cbica/brats2020/>
- [5] Ho, J., Jain, A., & Abbeel, P. (2020). Denoising Diffusion Probabilistic Models. *ArXiv preprint arXiv:2006.11239*. <https://arxiv.org/abs/2006.11239>
- [6] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015* (pp. 234–241). Springer. https://link.springer.com/chapter/10.1007/978-3-319-24574-4_28
- [7] Vinod, R. (2020). A detailed explanation of the Attention U-Net. Medium. <https://medium.com/data-science/a-detailed-explanation-of-the-attention-u-net-b371a5590831>
- [8] Latif, Z. (2024). Brain Tumor Segmentation Using U-Net [Kaggle Notebook]. Kaggle. <https://www.kaggle.com/code/zeeshanlatif/brain-tumor-segmentation-using-u-net/notebook>
- [9] prithivMLmods. (2025). Radiology-Infer-Mini [Computer software]. Hugging Face. <https://huggingface.co/prithivMLmods/Radiology-Infer-Mini>
- [10] Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., Fan, Y., Dang, K., Du, M., Ren, X., Men, R., Liu, D., Zhou, C., Zhou, J., & Lin, J. (2024, September 18). QWEN2-VL: Enhancing Vision-Language Model's perception of the world at any resolution. arXiv.org. <https://arxiv.org/abs/2409.12191>
- [11] Herencia Garcia del Castillo, M., Moya Garcia, R., Cerezo Mazón, M.J., Arriola Garcia, E., & Menéndez Fernández-Miranda, P (2025). Diffusion Models for Conditional MRI Generation. *ArXiv preprint arXiv:2502.18260*. <https://arxiv.org/abs/2502.18260>
- [12] Reimers, N., & Gurevych, I. (2019, August 27). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. arXiv.org. <https://arxiv.org/abs/1908.10084>